

```
!pip install d2l==0.17.0
!pip install tensorflow
import tensorflow as tf
from d2l import tensorflow as d2l
import matplotlib.pyplot as plt
import numpy as np
import time
```

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Requirement already satisfied: d2l==0.17.0 in
c:\users\ro517\anaconda3\lib\site-packages (0.17.0)
Requirement already satisfied: jupyter in c:\users\ro517\anaconda3\lib\site-
packages (from d2l==0.17.0) (1.1.1)
Requirement already satisfied: numpy in c:\users\ro517\anaconda3\lib\site-
packages (from d2l==0.17.0) (2.1.3)
Requirement already satisfied: matplotlib in
c:\users\ro517\anaconda3\lib\site-packages (from d2l==0.17.0) (3.10.0)
Requirement already satisfied: requests in c:\users\ro517\anaconda3\lib\site-
packages (from d2l==0.17.0) (2.32.3)
Requirement already satisfied: pandas in c:\users\ro517\anaconda3\lib\site-
packages (from d2l==0.17.0) (2.2.3)
Requirement already satisfied: notebook in c:\users\ro517\anaconda3\lib\site-
packages (from jupyter->d2l==0.17.0) (7.3.2)
Requirement already satisfied: jupyter-console in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter->d2l==0.17.0) (6.6.3)
Requirement already satisfied: nbconvert in c:\users\ro517\anaconda3\lib\site-
packages (from jupyter->d2l==0.17.0) (7.16.6)
Requirement already satisfied: ipykernel in c:\users\ro517\anaconda3\lib\site-
packages (from jupyter->d2l==0.17.0) (6.29.5)
Requirement already satisfied: ipywidgets in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter->d2l==0.17.0) (8.1.5)
Requirement already satisfied: jupyterlab in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter->d2l==0.17.0) (4.3.4)
Requirement already satisfied: comm>=0.1.1 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter-
>d2l==0.17.0) (0.2.1)
Requirement already satisfied: debugpy>=1.6.5 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter-
>d2l==0.17.0) (1.8.11)
Requirement already satisfied: ipython>=7.23.1 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter-
>d2l==0.17.0) (8.30.0)
Requirement already satisfied: jupyter-client>=6.1.12 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter-
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>d2l==0.17.0) (8.6.3)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (5.7.2)
Requirement already satisfied: matplotlib-inline>=0.1 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (0.1.6)
Requirement already satisfied: nest-asyncio in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (1.6.0)
Requirement already satisfied: packaging in c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (24.2)
Requirement already satisfied: psutil in c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (5.9.0)
Requirement already satisfied: pyzmq>=24 in c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (26.2.0)
Requirement already satisfied: tornado>=6.1 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (6.5.1)
Requirement already satisfied: traitlets>=5.4.0 in
c:\users\ro517\anaconda3\lib\site-packages (from ipykernel->jupyter->d2l==0.17.0) (5.14.3)
Requirement already satisfied: decorator in c:\users\ro517\anaconda3\lib\site-packages (from ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (5.1.1)
Requirement already satisfied: jedi>=0.16 in
c:\users\ro517\anaconda3\lib\site-packages (from ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (0.19.2)
Requirement already satisfied: prompt-toolkit<3.1.0,>=3.0.41 in
c:\users\ro517\anaconda3\lib\site-packages (from ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (3.0.43)
Requirement already satisfied: pygments>=2.4.0 in
c:\users\ro517\anaconda3\lib\site-packages (from ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (2.19.1)
Requirement already satisfied: stack-data in
c:\users\ro517\anaconda3\lib\site-packages (from ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (0.2.0)
Requirement already satisfied: colorama in c:\users\ro517\anaconda3\lib\site-packages (from ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (0.4.6)
Requirement already satisfied: wcwidth in c:\users\ro517\anaconda3\lib\site-packages (from prompt-toolkit<3.1.0,>=3.0.41->ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (0.2.5)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
c:\users\ro517\anaconda3\lib\site-packages (from jedi>=0.16->ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (0.8.4)
Requirement already satisfied: python-dateutil>=2.8.2 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-client>=6.1.12-

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>ipykernel->jupyter->d2l==0.17.0) (2.9.0.post0)
Requirement already satisfied: platformdirs>=2.5 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-core!=5.0.*,>=4.12-
>ipykernel->jupyter->d2l==0.17.0) (4.3.7)
Requirement already satisfied: pywin32>=300 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-core!=5.0.*,>=4.12-
>ipykernel->jupyter->d2l==0.17.0) (308)
Requirement already satisfied: six>=1.5 in c:\users\ro517\anaconda3\lib\site-
packages (from python-dateutil>=2.8.2->jupyter-client>=6.1.12->ipykernel-
>jupyter->d2l==0.17.0) (1.17.0)
Requirement already satisfied: widgetsnbextension~=4.0.12 in
c:\users\ro517\anaconda3\lib\site-packages (from ipywidgets->jupyter-
>d2l==0.17.0) (4.0.13)
Requirement already satisfied: jupyterlab_widgets~=3.0.12 in
c:\users\ro517\anaconda3\lib\site-packages (from ipywidgets->jupyter-
>d2l==0.17.0) (3.0.13)
Requirement already satisfied: async-lru>=1.0.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (2.0.4)
Requirement already satisfied: httpx>=0.25.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (0.28.1)
Requirement already satisfied: jinja2>=3.0.3 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (3.1.6)
Requirement already satisfied: jupyter-lsp>=2.0.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (2.2.5)
Requirement already satisfied: jupyter-server<3,>=2.4.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (2.15.0)
Requirement already satisfied: jupyterlab-server<3,>=2.27.1 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (2.27.3)
Requirement already satisfied: notebook-shim>=0.2 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (0.2.4)
Requirement already satisfied: setuptools>=40.8.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab->jupyter-
>d2l==0.17.0) (72.1.0)
Requirement already satisfied: anyio>=3.1.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0-
>jupyterlab->jupyter->d2l==0.17.0) (4.7.0)
Requirement already satisfied: argon2-cffi>=21.1 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0-
>jupyterlab->jupyter->d2l==0.17.0) (21.3.0)
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Requirement already satisfied: jupyter-events>=0.11.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (0.12.0)

Requirement already satisfied: jupyter-server-terminals>=0.4.4 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (0.5.3)

Requirement already satisfied: nbformat>=5.3.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (5.10.4)

Requirement already satisfied: overrides>=5.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (7.4.0)

Requirement already satisfied: prometheus-client>=0.9 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (0.21.1)

Requirement already satisfied: pywinpty>=2.0.1 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (2.0.15)

Requirement already satisfied: send2trash>=1.8.2 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (1.8.2)

Requirement already satisfied: terminado>=0.8.3 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (0.17.1)

Requirement already satisfied: websocket-client>=1.7 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (1.8.0)

Requirement already satisfied: babel>=2.10 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter->d2l==0.17.0) (2.16.0)

Requirement already satisfied: json5>=0.9.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter->d2l==0.17.0) (0.9.25)

Requirement already satisfied: jsonschema>=4.18.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter->d2l==0.17.0) (4.23.0)

Requirement already satisfied: idna>=2.8 in c:\users\ro517\anaconda3\lib\site-packages (from anyio>=3.1.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (3.7)

Requirement already satisfied: sniffio>=1.1 in
c:\users\ro517\anaconda3\lib\site-packages (from anyio>=3.1.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (1.3.0)

Requirement already satisfied: argon2-cffi-bindings in
c:\users\ro517\anaconda3\lib\site-packages (from argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (21.2.0)

Requirement already satisfied: certifi in c:\users\ro517\anaconda3\lib\site-

packages (from httpx>=0.25.0->jupyterlab->jupyter->d2l==0.17.0) (2025.10.5)
Requirement already satisfied: httpcore==1.* in
c:\users\ro517\anaconda3\lib\site-packages (from httpx>=0.25.0->jupyterlab->jupyter->d2l==0.17.0) (1.0.9)
Requirement already satisfied: h11>=0.16 in c:\users\ro517\anaconda3\lib\site-packages (from httpcore==1.*->httpx>=0.25.0->jupyterlab->jupyter->d2l==0.17.0) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jinja2>=3.0.3->jupyterlab->jupyter->d2l==0.17.0) (3.0.2)
Requirement already satisfied: attrs>=22.2.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter->d2l==0.17.0) (24.3.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in
c:\users\ro517\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter->d2l==0.17.0) (2023.7.1)
Requirement already satisfied: referencing>=0.28.4 in
c:\users\ro517\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter->d2l==0.17.0) (0.30.2)
Requirement already satisfied: rpds-py>=0.7.1 in
c:\users\ro517\anaconda3\lib\site-packages (from jsonschema>=4.18.0->jupyterlab-server<3,>=2.27.1->jupyterlab->jupyter->d2l==0.17.0) (0.22.3)
Requirement already satisfied: python-json-logger>=2.0.4 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (3.2.1)
Requirement already satisfied: pyyaml>=5.3 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (6.0.2)
Requirement already satisfied: rfc3339-validator in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (0.1.4)
Requirement already satisfied: rfc3986-validator>=0.1.1 in
c:\users\ro517\anaconda3\lib\site-packages (from jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (0.1.1)
Requirement already satisfied: fqdn in c:\users\ro517\anaconda3\lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (1.5.1)
Requirement already satisfied: isoduration in
c:\users\ro517\anaconda3\lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (20.11.0)
Requirement already satisfied: jsonpointer>1.13 in
c:\users\ro517\anaconda3\lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (2.1)
Requirement already satisfied: uri-template in

c:\users\ro517\anaconda3\lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (1.3.0)
Requirement already satisfied: webcolors>=24.6.0 in
c:\users\ro517\anaconda3\lib\site-packages (from jsonschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (25.10.0)
Requirement already satisfied: beautifulsoup4 in
c:\users\ro517\anaconda3\lib\site-packages (from nbconvert->jupyter->d2l==0.17.0) (4.12.3)
Requirement already satisfied: bleach!=5.0.0 in
c:\users\ro517\anaconda3\lib\site-packages (from bleach[css]!=5.0.0->nbconvert->jupyter->d2l==0.17.0) (6.2.0)
Requirement already satisfied: defusedxml in
c:\users\ro517\anaconda3\lib\site-packages (from nbconvert->jupyter->d2l==0.17.0) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in
c:\users\ro517\anaconda3\lib\site-packages (from nbconvert->jupyter->d2l==0.17.0) (0.3.0)
Requirement already satisfied: mistune<4,>=2.0.3 in
c:\users\ro517\anaconda3\lib\site-packages (from nbconvert->jupyter->d2l==0.17.0) (3.1.2)
Requirement already satisfied: nbclient>=0.5.0 in
c:\users\ro517\anaconda3\lib\site-packages (from nbconvert->jupyter->d2l==0.17.0) (0.10.2)
Requirement already satisfied: pandocfilters>=1.4.1 in
c:\users\ro517\anaconda3\lib\site-packages (from nbconvert->jupyter->d2l==0.17.0) (1.5.0)
Requirement already satisfied: webencodings in
c:\users\ro517\anaconda3\lib\site-packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert->jupyter->d2l==0.17.0) (0.5.1)
Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
c:\users\ro517\anaconda3\lib\site-packages (from bleach[css]!=5.0.0->nbconvert->jupyter->d2l==0.17.0) (1.4.0)
Requirement already satisfied: fastjsonschema>=2.15 in
c:\users\ro517\anaconda3\lib\site-packages (from nbformat>=5.3.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (2.20.0)
Requirement already satisfied: charset-normalizer<4,>=2 in
c:\users\ro517\anaconda3\lib\site-packages (from requests->d2l==0.17.0) (3.3.2)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\ro517\anaconda3\lib\site-packages (from requests->d2l==0.17.0) (2.3.0)
Requirement already satisfied: cffi>=1.0.1 in
c:\users\ro517\anaconda3\lib\site-packages (from argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0)

(1.17.1)

Requirement already satisfied: pycparser in c:\users\ro517\anaconda3\lib\site-packages (from cffi>=1.0.1->argon2-cffi-bindings->argon2-cffi>=21.1->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (2.21)

Requirement already satisfied: soupsieve>1.2 in c:\users\ro517\anaconda3\lib\site-packages (from beautifulsoup4->nbconvert->jupyter->d2l==0.17.0) (2.5)

Requirement already satisfied: arrow>=0.15.0 in c:\users\ro517\anaconda3\lib\site-packages (from isoduration->jschema[format-nongpl]>=4.18.0->jupyter-events>=0.11.0->jupyter-server<3,>=2.4.0->jupyterlab->jupyter->d2l==0.17.0) (1.3.0)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\ro517\anaconda3\lib\site-packages (from matplotlib->d2l==0.17.0) (1.3.1)

Requirement already satisfied: cyclor>=0.10 in c:\users\ro517\anaconda3\lib\site-packages (from matplotlib->d2l==0.17.0) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\ro517\anaconda3\lib\site-packages (from matplotlib->d2l==0.17.0) (4.55.3)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\ro517\anaconda3\lib\site-packages (from matplotlib->d2l==0.17.0) (1.4.8)

Requirement already satisfied: pillow>=8 in c:\users\ro517\anaconda3\lib\site-packages (from matplotlib->d2l==0.17.0) (11.1.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\ro517\anaconda3\lib\site-packages (from matplotlib->d2l==0.17.0) (3.2.0)

Requirement already satisfied: pytz>=2020.1 in c:\users\ro517\anaconda3\lib\site-packages (from pandas->d2l==0.17.0) (2024.1)

Requirement already satisfied: tzdata>=2022.7 in c:\users\ro517\anaconda3\lib\site-packages (from pandas->d2l==0.17.0) (2025.2)

Requirement already satisfied: executing in c:\users\ro517\anaconda3\lib\site-packages (from stack-data->ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (0.8.3)

Requirement already satisfied: asttokens in c:\users\ro517\anaconda3\lib\site-packages (from stack-data->ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (3.0.0)

Requirement already satisfied: pure-eval in c:\users\ro517\anaconda3\lib\site-packages (from stack-data->ipython>=7.23.1->ipykernel->jupyter->d2l==0.17.0) (0.2.2)

Collecting weasyprint

Downloading weasyprint-66.0-py3-none-any.whl.metadata (3.7 kB)

Collecting pydyf>=0.11.0 (from weasyprint)

Downloading pydyf-0.11.0-py3-none-any.whl.metadata (2.5 kB)

Requirement already satisfied: cffi>=0.6 in c:\users\ro517\anaconda3\lib\site-

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packages (from weasyprint) (1.17.1)
Collecting tinyhtml5>=2.0.0b1 (from weasyprint)
  Downloading tinyhtml5-2.0.0-py3-none-any.whl.metadata (2.9 kB)
Requirement already satisfied: tinycss2>=1.4.0 in
c:\users\ro517\anaconda3\lib\site-packages (from weasyprint) (1.4.0)
Collecting cssselect2>=0.8.0 (from weasyprint)
  Downloading cssselect2-0.8.0-py3-none-any.whl.metadata (2.9 kB)
Collecting Pyphen>=0.9.1 (from weasyprint)
  Downloading pyphen-0.17.2-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: Pillow>=9.1.0 in
c:\users\ro517\anaconda3\lib\site-packages (from weasyprint) (11.1.0)
Requirement already satisfied: fonttools>=4.0.0 in
c:\users\ro517\anaconda3\lib\site-packages (from fonttools[woff]>=4.0.0-
>weasyprint) (4.55.3)
Requirement already satisfied: pycparser in c:\users\ro517\anaconda3\lib\site-
packages (from cffi>=0.6->weasyprint) (2.21)
Requirement already satisfied: webencodings in
c:\users\ro517\anaconda3\lib\site-packages (from cssselect2>=0.8.0-
>weasyprint) (0.5.1)
Collecting zopfli>=0.1.4 (from fonttools[woff]>=4.0.0->weasyprint)
  Downloading zopfli-0.4.0-cp310-abi3-win_amd64.whl.metadata (3.6 kB)
Requirement already satisfied: brotli>=1.0.1 in
c:\users\ro517\anaconda3\lib\site-packages (from fonttools[woff]>=4.0.0-
>weasyprint) (1.0.9)
Downloading weasyprint-66.0-py3-none-any.whl (301 kB)
Downloading cssselect2-0.8.0-py3-none-any.whl (15 kB)
Downloading pydyf-0.11.0-py3-none-any.whl (8.1 kB)
Downloading pyphen-0.17.2-py3-none-any.whl (2.1 MB)
----- 0.0/2.1 MB ? eta -:--:--
----- 0.3/2.1 MB ? eta -:--:--
----- 1.0/2.1 MB 3.3 MB/s eta 0:00:01
----- 2.1/2.1 MB 3.6 MB/s eta 0:00:00
Downloading tinyhtml5-2.0.0-py3-none-any.whl (39 kB)
Downloading zopfli-0.4.0-cp310-abi3-win_amd64.whl (100 kB)
Installing collected packages: zopfli, tinyhtml5, Pyphen, pydyf, cssselect2,
weasyprint

----- 2/6 [Pyphen]
----- 5/6 [weasyprint]
----- 5/6 [weasyprint]
----- 5/6 [weasyprint]
----- 6/6 [weasyprint]

Successfully installed Pyphen-0.17.2 cssselect2-0.8.0 pydyf-0.11.0 tinyhtml5-
2.0.0 weasyprint-66.0 zopfli-0.4.0
Requirement already satisfied: tensorflow in

```


c:\users\ro517\anaconda3\lib\site-packages (2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (2.3.1)
Requirement already satisfied: astunparse>=1.6.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (25.9.23)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (0.6.0)
Requirement already satisfied: google_pasta>=0.1.1 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (0.2.0)
Requirement already satisfied: libclang>=13.0.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (18.1.1)
Requirement already satisfied: opt_einsum>=2.3.2 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (3.4.0)
Requirement already satisfied: packaging in c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (24.2)
Requirement already satisfied: protobuf>=5.28.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (5.29.3)
Requirement already satisfied: requests<3,>=2.21.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (2.32.3)
Requirement already satisfied: setuptools in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (72.1.0)
Requirement already satisfied: six>=1.12.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: termcolor>=1.1.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (3.2.0)
Requirement already satisfied: typing_extensions>=3.6.6 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (4.12.2)
Requirement already satisfied: wrapt>=1.11.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (1.17.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (1.76.0)
Requirement already satisfied: tensorboard~=2.20.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (2.20.0)
Requirement already satisfied: keras>=3.10.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (3.12.0)
Requirement already satisfied: numpy>=1.26.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (2.1.3)
Requirement already satisfied: h5py>=3.11.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (3.12.1)
Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorflow) (0.5.3)
Requirement already satisfied: charset-normalizer<4,>=2 in
c:\users\ro517\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in
c:\users\ro517\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.7)

Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\ro517\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2.3.0)

Requirement already satisfied: certifi>=2017.4.17 in
c:\users\ro517\anaconda3\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (2025.10.5)

Requirement already satisfied: markdown>=2.6.8 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (3.8)

Requirement already satisfied: pillow in c:\users\ro517\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (11.1.0)

Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (0.7.2)

Requirement already satisfied: werkzeug>=1.0.1 in
c:\users\ro517\anaconda3\lib\site-packages (from tensorboard~=2.20.0->tensorflow) (3.1.3)

Requirement already satisfied: wheel<1.0,>=0.23.0 in
c:\users\ro517\anaconda3\lib\site-packages (from astunparse>=1.6.0->tensorflow) (0.45.1)

Requirement already satisfied: rich in c:\users\ro517\anaconda3\lib\site-packages (from keras>=3.10.0->tensorflow) (13.9.4)

Requirement already satisfied: namex in c:\users\ro517\anaconda3\lib\site-packages (from keras>=3.10.0->tensorflow) (0.1.0)

Requirement already satisfied: optree in c:\users\ro517\anaconda3\lib\site-packages (from keras>=3.10.0->tensorflow) (0.17.0)

Requirement already satisfied: MarkupSafe>=2.1.1 in
c:\users\ro517\anaconda3\lib\site-packages (from werkzeug>=1.0.1->tensorboard~=2.20.0->tensorflow) (3.0.2)

Requirement already satisfied: markdown-it-py>=2.2.0 in
c:\users\ro517\anaconda3\lib\site-packages (from rich->keras>=3.10.0->tensorflow) (2.2.0)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
c:\users\ro517\anaconda3\lib\site-packages (from rich->keras>=3.10.0->tensorflow) (2.19.1)

Requirement already satisfied: mdurl~=0.1 in
c:\users\ro517\anaconda3\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.10.0->tensorflow) (0.1.0)

```
def simplified_alexnet_224():  
    return tf.keras.models.Sequential([  
        # Минимальный сверточный блок
```

PYTHON

```

tf.keras.layers.Conv2D(16, 5, strides=2, activation='relu'),
tf.keras.layers.MaxPool2D(2),

# Еще один простой слой
tf.keras.layers.Conv2D(32, 3, activation='relu'),
tf.keras.layers.MaxPool2D(2),

# Финальные слои - минимальные
tf.keras.layers.Flatten(),
tf.keras.layers.Dense(64, activation='relu'),
tf.keras.layers.Dense(10)
])

def alexnet_28x28():
    return tf.keras.models.Sequential([
        # Всего 2 сверточных слоя для 28x28
        tf.keras.layers.Conv2D(16, 3, activation='relu'),
        tf.keras.layers.MaxPool2D(2),

        tf.keras.layers.Conv2D(32, 3, activation='relu'),
        tf.keras.layers.MaxPool2D(2),

        # Минимальные полносвязные слои
        tf.keras.layers.Flatten(),
        tf.keras.layers.Dense(32, activation='relu'),
        tf.keras.layers.Dense(10)
    ])

# Функция для проверки размерностей
def check_output_shapes(model, input_shape):
    X = tf.random.uniform(input_shape)
    print("Проверка размерностей выходов слоев:")
    for layer in model.layers:
        X = layer(X)
        print(f"{layer.__class__.__name__:20} output shape: {X.shape}")

```

```

# Функция для обучения и оценки модели
def train_and_evaluate(model, batch_size, resize=None, model_name=""):
    print(f"\n{'='*50}")
    print(f"Обучение модели: {model_name}")
    print(f"Batch size: {batch_size}, Resize: {resize}")
    print(f"{'='*50}")

```

PYTHON

```

# Загрузка данных
train_iter, test_iter = d2l.load_data_fashion_mnist(batch_size, resize=resize)

# Обучение модели
lr, num_epochs = 0.01, 10
d2l.train_ch6(model, train_iter, test_iter, num_epochs, lr, d2l.try_gpu())

    return model
print("Упрощенная AlexNet для 224x224")
model1 = simplified_alexnet_224()
model1.build((None, 224, 224, 1))
print("Архитектура модели:")
model1.summary()

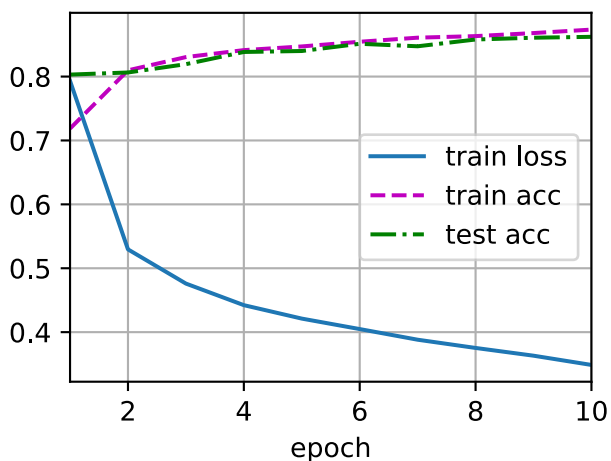
check_output_shapes(model1, (1, 224, 224, 1))
trained_model1 = train_and_evaluate(simplified_alexnet_224, batch_size=128, resize=224

```

```

loss 0.349, train acc 0.874, test acc 0.862
1412.0 examples/sec on /CPU:0

```



```

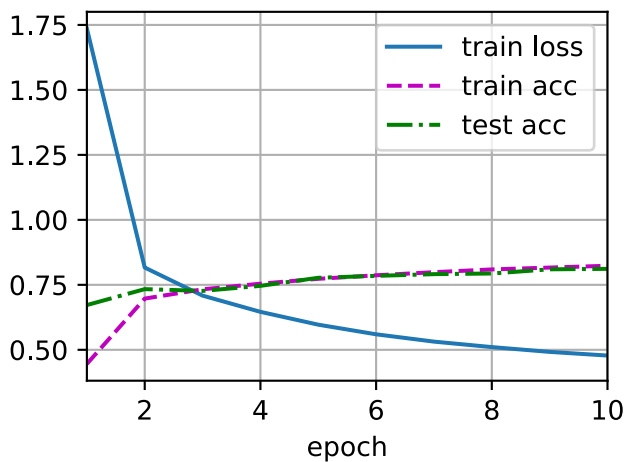
print("\n AlexNet адаптированная для 28x28")
model2 = alexnet_28x28()
model2.build((None, 28, 28, 1))
print("Архитектура модели:")
model2.summary()

check_output_shapes(model2, (1, 28, 28, 1))
trained_model2 = train_and_evaluate(alexnet_28x28, batch_size=128, resize=None, model_

```

PYTHON

loss 0.478, train acc 0.823, test acc 0.812
16344.4 examples/sec on /CPU:0



```
# Задание 3 - Исправленная версия
print("\n Влияние размера пакета на производительность")
batch_sizes = [64, 128, 256]

# Исправленная функция для AlexNet 28x28 с указанием input_shape
def alexnet_28x28_fixed():
    return tf.keras.models.Sequential([
        tf.keras.layers.Conv2D(16, 3, activation='relu', input_shape=(28, 28, 1)),
        tf.keras.layers.MaxPool2D(2),
        tf.keras.layers.Conv2D(32, 3, activation='relu'),
        tf.keras.layers.MaxPool2D(2),
        tf.keras.layers.Flatten(),
        tf.keras.layers.Dense(32, activation='relu'),
        tf.keras.layers.Dense(10)
    ])

for batch_size in batch_sizes:
    print(f"\nТестирование с batch_size = {batch_size}")

    # Мониторинг использования памяти
    print("Попытка использование памяти GPU до обучения:")
    get_ipython().system('nvidia-smi --query-gpu=memory.used --format=csv')
    train_iter, test_iter = d2l.load_data_fashion_mnist(batch_size, resize=None)

    lr, num_epochs = 0.01, 5
    # Передаем функцию
    d2l.train_ch6(alexnet_28x28_fixed, train_iter, test_iter, num_epochs, lr, d2l.try_
```

PYTHON

```
print("Использование памяти GPU после обучения:")  
get_ipython().system('nvidia-smi --query-gpu=memory.used --format=csv')
```

```
loss 0.725, train acc 0.733, test acc 0.730  
20550.1 examples/sec on /CPU:0  
Использование памяти GPU после обучения:  
memory.used [MiB]  
19 MiB
```

